LYRA RECORDER

Technical Datasheet · Version 1.0

Overview

Lyra Recorder is a professional Android audio capture application with direct and calibrated recording workflows, real-time signal visualization, optional signal processing and adaptive silence detection. The instrument is designed for practical relative monitoring rather than certified acoustic measurement.

Audio engine

Capture API	Android AudioRecord (Kotlin native layer)
Encoding API	MediaCodec + MediaMuxer for AAC
Sample rate	44,100 Hz
Bit depth	16-bit PCM
Channels	Mono
Internal processing	32-bit floating-point signal processing
Buffering	Device-dependent AudioRecord buffering

Signal chain

The application captures microphone input, processes the digital signal as needed and writes the selected recording format. Optional processing includes a high-pass filter, gain adjustment and limiter; silence detection is used for AUTO PAUSE. The live visualization includes spectrum, waveform and separate RMS/peak indications.

Input conversion	16-bit PCM to normalized floating-point signal
Optional HPF	High-pass filtering for low-frequency components
Silence detection	Signal-level evaluation for AUTO PAUSE
GAIN	User-adjustable signal gain
Optional LIMITER	Peak-protection stage, when enabled
Output	WAV writer or AAC encoder; optional monitoring path
Analysis	Spectrum, waveform, relative RMS and peak indication

Calibration

Calibrated operation evaluates both the ambient noise floor and a normal reference voice signal. The resulting profile supports relative meter behavior and AUTO PAUSE thresholds. AUTO PAUSE may be restricted when ambient noise prevents reliable silence detection. These reference measurements are device- and environment-dependent and do not represent certified SPL calibration.

Silence reference	Guided ambient-noise measurement
Voice reference	Guided normal-level voice measurement
Noise assessment	Used to determine whether automatic silence detection is viable
Profile validity	A new calibration is advisable when recording conditions change

Metering

Analog VU	Sustained relative RMS indication (RMS · REL)
Peak indication	Independent peak warning/reference (PK ····· FS)
Spectrum	Live frequency distribution, frequency axis extending to 20 kHz
Spectrum units	Frequency in Hz/kHz; relative signal-level indication
Waveform	Time-domain view on normalized full-scale reference (FS; +1 / 0 / -1)
Purpose	Relative operational guidance, not a laboratory-grade acoustic measurement

The meter indications describe operational signal behavior; exact thresholds and response characteristics depend on the active processing and implementation.

Output formats

WAV	PCM 16-bit, 44.1 kHz, mono, uncompressed
AAC	MPEG-4 AAC-LC, 256 kbit/s, 44.1 kHz, mono, .m4a container
File naming	lyra_<timestamp_ms>.<ext>
Markers	Associated per-recording note/marker data

Features

Recording modes	Direct capture and calibrated operation
Calibration-dependent controls	HPF, LIMITER, AUTO PAUSE
Markers	Per-recording markers with optional text note (up to 50 characters)
AUTO PAUSE	OFF, 1 s and 2 s silence settings
Monitor	Live monitoring via the Android audio system
Playback	Overview/detail waveforms, navigation controls and marker access
Interface	Portrait-oriented instrument-style interface with controlled proportions

Platform and display requirements

Minimum OS	Android 7.0 (API 24) or higher
Device class	Android smartphone, portrait orientation
Screen aspect ratio	19.5:9 or taller
Unsupported formats	Tablets and foldable devices are not supported
Display rationale	The original instrument geometry and visual proportions are deliberately preserved on supported devices
Microphone access	Required for audio recording
Background capture	Requires Android foreground-service capabilities and notification behavior when recording continues outside the foreground

Trial and licensing

Trial	21 days from initial server registration
Purchase model	One-time Google Play purchase after the trial
Validation	Server-assisted license/trial validation; offline operation is subject to periodic validation

Implementation

UI layer	Flutter / Dart
Native layer	Kotlin (Android)
Audio capture	Android AudioRecord
AAC encoding	Android MediaCodec / MediaMuxer